FUNCTORIALITY, BASE CHANGE COMPATIBILITY, AND DERIVED CATEGORY VERSION OF THE TRACE MAPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Editorial note: the author proves part (1) by two alternative routes (the construction in the proof of lemma-trace-map-exists, or the stalk characterization forcing functoriality) and displays neither verification, treating the check as immediate on stalks; in part (2) the commutativity is concluded from the stalk characterization of the trace maps together with the asserted fact that the base change map of lemma-lqf-base-change-f-shriek respects the descriptions of stalks, a compatibility invoked rather than re-derived in this proof.

Allowed standard prerequisites (editorial).

- Existence of the trace maps for a weighted locally quasi-finite morphism (Lemma lemma-trace-map-exists, canonical tag 0GKG). cited at proof lines 2215-2216. Its construction is the first route for part (1) and its proof supplies the stalk characterization of the trace maps used throughout.
- Base change property of weightings (More on Morphisms, Lemma more-morphisms-lemma-weighting-base-change, canonical tag 0F3B). cited at proof lines 2227-2228. w' = w composed with g' is a weighting of f' for the cartesian square, so Tr_{f', w', -} is defined.
- Base change for lower shriek along locally quasi-finite morphisms (Lemma lemma-lqf-base-change-f-shriek, canonical tag 0F5J). cited at proof lines 2246-2250. Gives the first equality in the left vertical identification of the diagram; its base change map respecting the stalk descriptions is the fact concluding the commutativity.

Source reference. Original lemma 0GKG: sources/stacks-chow/tags.txt, source lines 20686–20686. Cited at proof lines 2215-2216: the construction referenced by the first route for part (1); its proof also supplies the stalk characterization of the trace maps used in parts (1) and (2).

Source reference. Original lemma 0F3B: sources/stacks-chow/tags.txt, source lines 18861–18861. Cited at proof lines 2227-2228: w' = w composed with g' is a weighting of f' for the cartesian square, so the trace maps for (f', w') are defined.

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Source reference. Original lemma 0F5J: sources/stacks-chow/tags.txt, source lines 18939–18939. Cited at proof lines 2246-2250: supplies the first equality in the left vertical identification of the diagram, and its base change map respecting the stalk descriptions is the fact concluding commutativity.

Lemma 0.2. *Let $f : X \to Y$ be a locally quasi-finite morphism of schemes. Let $w : X \to \mathbf{Z}$ be a weighting of f. The trace maps constructed above have the following properties:*

1. *$Tr_{f,w,\mathcal{F}}$ is functorial in $\mathcal{F}$,*
2. *$Tr_{f,w,\mathcal{F}}$ is compatible with arbitrary base change,*
3. *given a ring Λ and K in $D(Y_{étale}, \Lambda)$ we obtain $Tr_{f,w,K} : f_!f^{-1}K \to K$ functorial in K and compatible with arbitrary base change.*

Proof. Part (1) either follows from the construction of the trace map in the proof of Lemma 0GKG or more simply because the characterization of the map forces it to be true on all stalks. Let

$$\begin{array}{ccc} X' & \xrightarrow[g']{} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a cartesian diagram of schemes. Then the function $w' = w \circ g' : X' \to \mathbf{Z}$ is a weighting of f' by More on Morphisms, Lemma 0F3B. Statement (2) means that the diagram

$$\begin{array}{ccc} g^{-1}f_!f^{-1}\mathcal{F} & \xrightarrow[g^{-1}\mathrm{Tr}_{f,w,\mathcal{F}}]{} & g^{-1}\mathcal{F} \\ \| & & \| \\ f'_!(f')^{-1}g^{-1}\mathcal{F} & \xrightarrow{\mathrm{Tr}_{f',w',g^{-1}\mathcal{F}}} & g^{-1}\mathcal{F} \end{array}$$

is commutative where the left vertical equality is given by

$$g^{-1}f_!f^{-1}\mathcal{F} = f'_!(g')^{-1}f^{-1}\mathcal{F} = f'_!(f')^{-1}g^{-1}\mathcal{F}$$

with first equality sign given by Lemma 0F5J (base change for lower shriek). The commutativity of this diagram follows from the characterization of the action of our trace maps on stalks and the fact that the base change map of Lemma 0F5J respects the descriptions of stalks.

Given parts (1) and (2), part (3) follows as the functors $f^{-1} : D(Y_{étale}, \Lambda) \to D(X_{étale}, \Lambda)$ and $f_! : D(X_{étale}, \Lambda) \to D(Y_{étale}, \Lambda)$ are obtained by applying f^{-1} and $f_!$ to any complexes of modules representing the objects in question. □